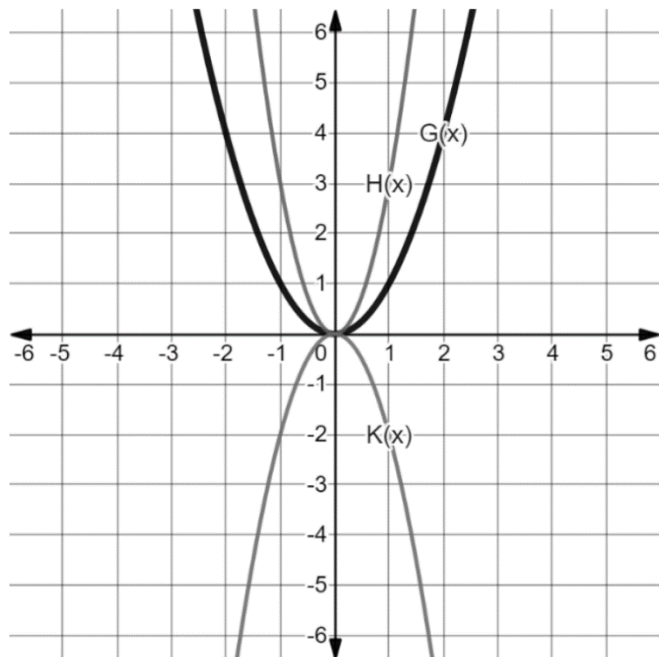


The functions $G(x)$, $H(x)$, and $K(x)$ are shown on the graph, where $G(x) = x^2$, $H(x) = 3x^2$, and $K(x) = -2x^2$. The table of values for each function are also shown.



$y = G(x)$		$y = H(x)$		$y = K(x)$	
x	y	x	y	x	y
-3	9	-3	27	-3	-18
-2	4	-2	12	-2	-8
-1	1	-1	3	-1	-2
0	0	0	0	0	0
1	1	1	3	1	-2
2	4	2	12	2	-8
3	9	3	27	3	-18

Complete the statements below.

1. The coefficient of $G(x) = x^2$ is **1**.
2. The coefficient of $H(x) = 3x^2$ is **3**.
3. The y -values of $H(x)$ can be found by multiplying the y -values of $G(x)$ by **3**.
4. The coefficient of $K(x) = -2x^2$ is **-2**.
5. The y -values of $K(x)$ can be found by multiplying the y -values of $G(x)$ by **-2**.

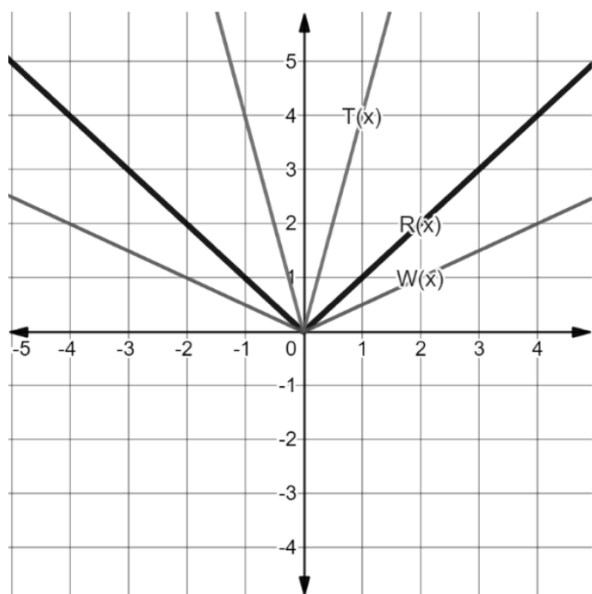
The table of values of the function $y = A(x)$ is shown.

x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

Complete the sentence to describe the effect on the x -values for the function $y = A(3x)$.

6. The x -values for $A(x)$ will be
☐ multiplied
☒ divided
 by 3.

The functions $R(x)$, $T(x)$, and $W(x)$ are shown on the graph, where $R(x) = |x|$, $T(x) = |4x|$, and $W(x) = |0.5x|$. The table of values for each function are also shown.



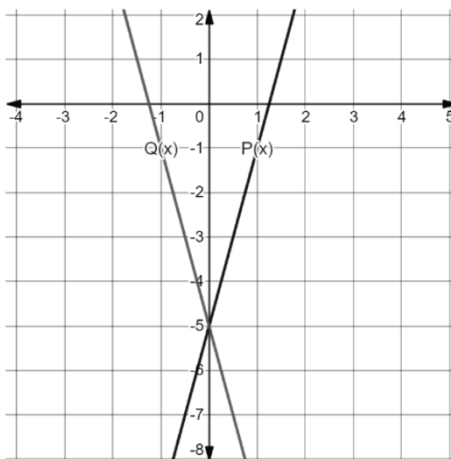
$y = R(x)$		$y = T(x)$		$y = W(x)$	
x	y	x	y	x	y
-2	2	$-\frac{1}{2}$	2	-4	2
-1	1	$-\frac{1}{4}$	1	-2	1
0	0	0	0	0	0
1	1	$\frac{1}{4}$	1	2	1
2	2	$\frac{1}{2}$	2	4	2

Complete the statements below.

7. The x -values of $T(x)$ can be found by ☐ multiplying ☒ dividing the x -values of $R(x)$ by 4.
8. The x -values of $W(x)$ can be found by ☐ multiplying ☒ dividing the x -values of $R(x)$ by $\frac{1}{2}$ or multiply by 2.

The functions $P(x)$ and $Q(x)$ are shown on the graph where $Q(x) = P(-x)$. The table of values for each function are also shown.

$y = P(x)$		$y = Q(x)$	
x	y	x	y
-2	-13	-2	3
-1	-9	-1	-1
0	-5	0	-5
1	-1	1	-9
2	3	2	-13



9. What is the effect on the table of values when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?
 Explanations may vary. The y -values changed to the opposite x -value.
10. What is the effect on the graph when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?
 reflects over the y -axis